

DHEERAJ KARWASRA

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SUMMARY

M.Sc. Computer Science student at RPTU Kaiserslautern with hands-on experience building scalable, fault-tolerant full-stack applications, AI pipelines, and distributed microservices on AWS, with strong foundations in data structures, algorithms, object-oriented design, operating systems, and complexity analysis.

WORK EXPERIENCE

Fraunhofer Software Developer – Capstone Project

Sep 2025 - Feb 2026

- Engineered a multi-agent RAG pipeline (CrewAI, LangChain, FastAPI) eliminating 80% of manual document handling across 3 departments via LLM-powered extraction, retrieval reranking, and prompt optimization.
- Designed an orchestration layer with dynamic pipeline switching based on document type, anomaly detection, and self-healing fallbacks; maintained 100% data integrity and cut downtime 9x.
- Delivered a full-stack generative AI prototype (React, FastAPI, PostgreSQL) replacing a 6-step manual requirements workflow with a single-click interface, led code reviews and authored technical documentation.

Highradius Software Developer Intern

Jan 2022 - Apr 2022

- Built and deployed an ML model (Random Forest + XGBoost) predicting invoice payment dates at 89% accuracy, improving cash-flow forecasting by 19% for enterprise B2B customers.

PROJECTS

Gatekeeper – Zero-Trust Reverse Proxy Gateway (Github)

Feb 2026 - Apr 2026

- Built a scalable zero-trust reverse proxy gateway (5 microservices) that authenticates, authorizes, and audits every request replacing VPN-based access and giving internal users secure access to services.
- Implements RS256 JWT auth, Redis-backed session revocation, mTLS encryption, RBAC policy engine, and a real-time monitoring dashboard with policy simulation sandbox following clean code design patterns.
- Tech: Python, FastAPI, React, TypeScript, Redis, PostgreSQL, Docker, OAuth 2.0.

ESN Event Management Platform (Github)

Aug 2025 - Nov 2025

- Built a scalable ESN event platform serving 300+ concurrent users, replacing fragmented Google Forms with automated waitlists, real-time registration management, and role-based access control.
- Optimized GraphQL API using DataLoader batching to reduce database roundtrips by 70 %, improving query complexity from $O(N)$ to $O(1)$ with Redis caching and Docker-based auto-scaling for high-traffic spikes.
- Tech: React 19, NestJS, GraphQL, Apollo, PostgreSQL, Prisma, Redis, Auth0, Docker, Nginx

EDUCATION

M.Sc. Computer Science Specialization – Software Engineering & Intelligent Systems

Apr 2024 - Present

RPTU Kaiserslautern, Germany

Bachelor of Engineering in Computer Science

Jul 2019 - Jun 2023

Chandigarh University, India

SKILLS

- Core CS:** Data Structures & Algorithms, OOP, Object-Oriented Design, Operating Systems, System Design, Complexity Analysis, Problem Solving.
- Languages:** Python, TypeScript, JavaScript, C++, SQL, Java
- Backend & Cloud:** FastAPI, Node.js, Express.js, GraphQL, AWS (EC2, S3, Lambda, RDS), Docker, CI/CD, Git
- Frontend:** React.js, Redux, Tailwind CSS, HTML/CSS
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, GraphDB(Neo4j)
- GenAI/AI:** LangChain, LangGraph, CrewAI, MCP, RAG, Prompt Engineering
- Soft Skills:** Team Collaboration, Ownership, Adaptability, Time Management, Analytical Thinking

CERTIFICATIONS

- Microsoft AI & ML Engineering Professional Certificate – Microsoft / Coursera
- Building AI Agents and Agentic Workflows Specialization – IBM / Coursera, 2025